📘 ALTIS - Phase 2:

Database Implementation

✅ Objective

The objective of Phase 2 is to implement the normalized database schema designed in Phase 1 into SQL Server. This includes:

- Creating normalized tables with appropriate constraints
- Inserting realistic data
- Writing and testing CRUD operations (Create, Read, Update, Delete)
- Enforcing data integrity through constraints and relationships

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🧩 1. Schema Implementation

The schema consists of 13 normalized tables in 3NF. Each table was created with:

- Primary Keys
- Foreign Keys
- Unique Constraints
- Check Constraints where necessary

Refer to the SQL DDL scripts for the full CREATE TABLE statements.

Key Entities:

Table Name	Description
ADMINS	Admin staff of ALTIS
STUDENTS	Student information and login details
FACULTY	Faculty and instructors
COURSES	List of academic courses
SECTIONS	Class sectioning for grouping students
COURSE_ASSIGNMENTS	Mapping of courses to faculty, sections, and semester
CLASSROOMS	Virtual classrooms for each course-section-faculty
ENROLLMENTS	Students enrolled in classrooms
ASSIGNMENTS	Assignments linked to classrooms
SUBMISSIONS	Student assignment submissions
ATTENDANCE	Student attendance per classroom
MARKS	Marks awarded for assignments
RESULTS	Final GPA and remarks per student

📥 2. Data Insertion

Each table was populated with realistic data including Pakistani names such as:

- Students: Taimur, Fatima, Ayesha, Usman, Raheem, etc.
- Faculty: Dr. Haseeb, Prof. Ahmad, etc.
- Courses: Computer Science, Mathematics, English, etc.
- Sections: A1, B1, C1…

Minimum 13 records per table were inserted for representative testing.

Refer to SQL INSERT statements file for full sample data population.

🔄 3. CRUD Operations

All tables were tested with:

🔹 a. INSERT

Used to add records to each table using both raw SQL and stored procedures.

Example:

```
INSERT INTO STUDENTS (StudentID, FirstName, LastName, ...)
VALUES (101, 'Taimur', 'Ali', ...);
```

🔹 b. SELECT (Read)

Multiple read queries were tested, including JOINs across tables.

Example:

```
SELECT S.FirstName, A.Title, M.Score
FROM MARKS M
```

```
JOIN STUDENTS S ON M.StudentID = S.StudentID
JOIN ASSIGNMENTS A ON M.AssignmentID = A.AssignmentID;
```

🔹 **c. UPDATE**

Tested updates on due dates, GPA, and user information.

Example:

```
UPDATE ASSIGNMENTS
SET DueDate = '2025-06-30'
WHERE AssignmentID = 2001;
```

🔹 **d. DELETE**

Cascading deletions tested via stored procedure (manual cascade to child tables first due to FK constraints).

Example:

```
DELETE FROM ENROLLMENTS WHERE StudentID = 101;
DELETE FROM STUDENTS WHERE StudentID = 101;
```

🔐 4. Constraints and Data Integrity

- PRIMARY KEY constraints on all ID fields
- FOREIGN KEY constraints for all relational mappings
- UNIQUE constraints on email addresses
- CHECK constraints on GPA (0.0–4.0), Score (0–100), and Attendance status ('Present'/'Absent')
- Referential integrity enforced strictly

🛠 5. Stored Procedures

Reusable procedures were created for:

- Adding students, course assignments
- Fetching student marks
- Updating due dates
- Deleting student (with child data cleanup)

Example:

```
EXEC AddStudent
 @StudentID = 121,
 @FirstName = 'Hassan',
 ...
```

✅ Summary

Phase 2 successfully implements a fully normalized and relational ALTIS database system on SQL Server. Data is structured, accessible, and modifiable through efficient SQL queries and stored procedures. The foundation is now ready for Phase 3 – UI development or integration.